

HAMZA KHAN

Senior Backend Engineer | Python · FastAPI · Kafka · AWS · GCP · Distributed Systems

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PROFESSIONAL SUMMARY

Backend engineer with 5 years of experience shipping production Python services at scale. Delivered a 30% reduction in database response time, cut peak-traffic queue latency by 40%, and built AI-powered pipelines now serving millions of users. Specialises in FastAPI microservices, Kafka streaming, Redis, AWS, and GCP, with a track record of translating complex distributed-systems problems into measurable reliability and cost wins.

TECHNICAL SKILLS

Languages: Python (primary), JavaScript

Backend: FastAPI, Django, Flask, REST APIs, Microservices, Async Programming

Messaging & Data: Kafka, Redis, Celery, MySQL, MongoDB, PostgreSQL

Cloud & DevOps: AWS (EC2, S3, SQS, Lambda), GCP (Kubernetes, Cloud Build, App Engine, Cloud Run), Docker, GitHub Actions, Jenkins, CI/CD

Architecture: Distributed Systems, Event-Driven Design, System Design, Scalable Services

PROFESSIONAL EXPERIENCE

Techolution

Jun 2025 – Present

Senior Software Engineer

Remote

- Architected Python microservices for a high-scale audio streaming and podcast platform handling 50M+ monthly active users across 850+ radio stations.
- Engineered an AI-powered document ingestion pipeline (Python, Celery, Redis, multi-agent LLM) that reduced manual processing time by ~70%, onboarding 10K+ documents/month.
- Resolved asynchronous context-propagation bugs that had caused ~15% metadata inconsistency across real-time podcast pipelines, bringing error rate to <1%.
- Built Redis-backed failure-tracking for LLM API workflows, cutting undetected task failures by 90% and reducing on-call alerts by ~30%.
- Optimised critical SQL queries and backend workflows, contributing to a 30% reduction in average database response time across core services.
- Designed and maintained Kafka streaming pipelines processing 100K+ real-time audio events per hour with sub-second end-to-end latency.

Xempla - Enterprise Asset Management Platform

Nov 2021 – Jun 2025

Software Engineer III

Remote (Australia)

- Built and scaled backend services and large data pipelines for enterprise analytics workloads across 200+ client sites globally.
- Redesigned pipeline architecture to reduce infrastructure load by 35%, directly lowering cloud compute costs by an estimated \$80K/year.
- Delivered analytics systems and API integrations that accelerated new product-line launch by 3 months and onboarded 15+ enterprise customers in year one.
- Diagnosed and resolved 30+ complex production incidents, achieving a 99.7% service uptime SLA across all managed pipelines.
- Mentored 3 junior engineers; introduced code-review standards that cut post-deployment bug reports by ~25%.

WatchMyDC Analytics

Mar 2021 – Nov 2021

Software Engineer

Remote (Finland)

- Implemented observability tooling (metrics, alerts, dashboards) that improved mean time to detection (MTTD) for system incidents from ~45 min to <10 min.
- Developed scalable Django REST APIs for CRM systems supporting 5K+ daily active users, achieving <200 ms p95 response time.

Croudit

Full Stack Developer

Oct 2020 – Mar 2021

Remote

- Designed server-side architecture in Python (Django and Flask), delivering 4 backend modules used across 2 product lines.
- Built infrastructure tooling that automated 3 manual deployment steps, saving the team ~5 hours/week.

KEY ACHIEVEMENTS

- 30% reduction in database response time through systematic query optimisation and index tuning.
- 40% improvement in peak-traffic throughput via queue-based architecture; zero SLA breaches during 3x traffic spikes.
- 70% decrease in manual processing overhead for AI document ingestion pipeline, scaling to 10K+ documents/month.
- \$80K/year infrastructure cost saving from pipeline efficiency improvements at Xempla.
- Built kynstro.com, an AI interview platform, end-to-end as a solo backend engineer.

EDUCATION

Bachelor in Computer Application, Integral University, Lucknow

2019

CGPA: 8.9 / 10